

ORIGINAL ARTICLE

Comparison of cauliflower–insect–fungus interactions and pesticides for cabbage root fly control

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Abstract Cabbage root fly (*Delia radicum* L.) control represents a major challenge in brassica production, therefore different management strategies for its control were tested in conventionally managed open field cauliflower production. Strategies included treatments with low-risk methods such as nitrogen lime, the insecticide spinosad and the *Beauveria bassiana* ATCC 74040-based biopesticide Naturalis. Their effects were compared with treatments based on nonformulated fungal species *Metarhizium brunneum*, *B. bassiana*, *Clonostachys solani*, *Trichoderma atroviride*, *T. koningiopsis*, and *T. gamsii* and commercial insecticides λ -cyhalothrin and thiamethoxam. Spinosad and thiamethoxam were pipetted to individual plants before transplanting; λ -cyhalothrin was sprayed after transplanting; nitrogen lime was applied at first hoeing. Nonformulated fungi were delivered onto cauliflower plantlets' roots as a single pretransplantation inoculation. The cabbage root fly population dynamics exhibited a strong spatiotemporal variation. The lowest number of cabbage root fly pupae recovered from cauliflower roots in the field experiments was recorded in plants treated with spinosad (significant reduction), followed by Naturalis and one of the tested *M. brunneum* strains (nonsignificant reduction). Significantly more pupae were counted in the nitrogen lime treatment. The field experiments showed that a single drench of cauliflower plantlets with spinosad offered consistent and enduring cabbage root fly control. Naturalis and nonformulated fungal isolates did not decrease cabbage root fly pressure significantly, apparently due to lack of statistical power. The implications of the substantial intra- and inter-annual pest pressure variation and the benefits of using single plant treatments are discussed, and recommendations for improvement of rhizosphere-competence utilizing biological control strategies provided.

Key words biological control; cabbage maggot; *Delia radicum*; entomopathogenic fungi; plant–insect–microbe interactions; rhizosphere competence

Introduction

Low-risk plant protection methods that avoid carcinogenic, mutagenic, or persistent (Regulation 1107/2009;

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European Parliament, 2009a) ingredients pose lesser environmental and health-related risks. They could therefore present an important tool allowing sustainable intensification of agricultural production (Godfray *et al.*, 2010) in conventional, integrated, and organic production systems.

The cabbage maggot or cabbage root fly *Delia radicum* L. (CRF) is one of the major soil pests in brassica production in the northern temperate region (Klingen *et al.*, 2002). Produce losses of up to 100% (Ferry *et al.*, 2009)

occur and young plants in particular are at risk (Bligaard, 1999). Direct damage occurs if root crops such as rutabaga or turnips are infested and even slight feeding damage becomes economically significant (Bruck *et al.*, 2005). Indirect damage occurs when larvae feed on nonmarketed plant parts such as roots in cauliflower production. According to Vrabl (1986), indirect damage in brassica, that is, low pest pressure measured in numbers of egg per plant per day, can be tolerated. For example, only young cauliflower transplants require insecticide treatment to achieve acceptable control of CRF (Bligaard, 1999).

Since organophosphate and carbamate insecticides and neonicotinoid seed treatments can no more be used in the EU, producers rely on the use of pyrethroid and spinosad-based insecticides for the control of CRF in brassica. However, other low-risk methods have been investigated, including nitrogen lime fertilizers that affect weeds, pathogenic fungi and insects on the basis of Cyanamid production and straw mulch soil cover (Herbst *et al.*, 2017). Entomopathogenic fungi or biopesticide formulations based, for example, on species of genera *Beauveria* and *Metarhizium* (henceforth abbreviated as EPF), are considered to be environmentally friendly (Vänninen *et al.*, 1999; Bruck *et al.*, 2005; Chandler & Davidson, 2005; Razinger *et al.*, 2014a,b; Myrand *et al.*, 2015; Lacey *et al.*, 2015) although inundative applications of products render biopesticide-based management expensive. Accordingly, one aim of this study was to test EPF and other potentially plant growth promoting fungi whether they are able to colonize brassica roots and have the ability to achieve high inoculum pressures without repeated application. This strategy implies that EPF flourishing in the rhizosphere or on rhizosphere deposits could protect plants from insect pests in mutualistic interactions as already hypothesized by Vega *et al.* (2009).

Reduction of dependence on pesticides is an integral part of the European Union's agenda for agriculture (Directive 2009/128/EC; European Parliament, 2009b). A key element in reaching this goal is to promote Integrated Pest Management (IPM) strategies. In order to address this, a working group within the European Project PURE (Pesticide Use-and-Risk reduction in European farming systems with Integrated Pest Management, <http://www.pure-ipm.eu>) has aimed at evaluating various IPM tools for their ability to reduce pesticide-dependence in cauliflower production. Two studies to evaluate various EPF for CRF control in cauliflower performed in laboratory and glasshouse settings (Razinger *et al.*, 2014a,b), as well as a similar study performed on experimental stations in broccoli (Herbst *et al.*, 2017), have been published from this working group. In this study, the results of field evaluation of several low-risk methods are presented, together

with a biological control strategy involving cauliflower plantlet inoculation with beneficial fungi (EPF and plant growth promoting fungi) as an IPM tool for *Delia radicum* management. Our aims were to compare CRF control efficacy of selected low-risk plant protection methods with the efficacy of selected unformulated fungal isolates. Experiments were done under field conditions in commercial farms that also used conventional insecticides over several seasons (2012–2014) under a variety of CRF pressures.

Materials and methods

Location and experimental set-up

The 2012 and 2013 experiments were performed with cauliflower cultivar “Fremont” in Kovor near Tržič, Slovenia (46°19'48"N, 14°16'41"E), with an intrarow distance of plants of 50 cm and an inter-row distance of 60 cm. The field of 0.59 ha was situated at 495 m asl. It was handled under a 4-year crop rotation regime involving cereals, a grass legume mixture, potatoes, and cauliflower. The experiment in 2014 was performed with cauliflower cultivar “Romanesco” in Šentjakob near Ljubljana, Slovenia (46°05'21"N, 14°34'22"E), with an intrarow distance of plants of 35 cm and an inter-row distance of 75 cm. The field of 2.76 ha was situated 278 m asl. It was handled under a 3-year crop rotation regime involving cereals, potatoes, and cauliflower. Both types of field experiments (setups 1 and 2; see below) were performed in the same field, under normal agronomic practice for commercial cauliflower production. Four-week-old cauliflower plantlets growing in planting trays were treated with spinosad, thiamethoxam, and formulated and nonformulated fungi 24 h before transplanting to the field. The plantlets were sprayed with insecticide λ -cyhalothrin after transplanting and lime nitrogen was incorporated into the soil at hoeing. The CRF egg laying was monitored throughout growth seasons 2011–2014 using 20 felt traps (Ateliers Olbis, Le Mont-sur-Lausanne, Switzerland) at each experimental site.

Experimental setup 1 for evaluating insecticides

Insecticides Laser 240 SC (a.i. spinosad, 22.75% w/w; Dow Agrosciences; dose: 5.76 mg a.i. in a volume of 3 mL per plant) and Actara 25 WG (a.i. thiamethoxam, 25% w/w; Syngenta; dose: 3.25 mg a.i. in a volume of 2 mL per plant) were pipetted to individual plants before transplanting. Insecticide Karate Zeon (a.i. λ -cyhalothrin, 22.8% w/w; Syngenta; dose: 150 mL/ha) was sprayed after transplanting. Nitrogen lime (PERLKA®, AlzChem;

Table 1 *Delia radicum* (CRF) management in different experiments and years.

Experiment	Date of exp.	CRF management	Dose [†]	Dates of treatment
Setup 1 and 2	2011	Start of CRF monitoring	–	–
Setup 1	17/07 2012	Control	–	No treatment
		Spinosad	5.76 mg	16/07
		λ-cyhalothrin	34.2 g/ha	23/07, 02/08
		Thiamethoxam	3.25 mg	16/07
	25/07 2013	Control	–	No treatment
		Spinosad	5.76 mg	24/07
		Nitrogen lime	500 kg/ha	08/08
	23/06 2014	Control	–	No treatment
		Spinosad	5.76 mg	22/06
		λ-cyhalothrin	34.2 g/ha	26/06, 07/07
		Thiamethoxam	3.25 mg	22/06
Setup 2	17/07 2012	Control	–	No treatment
		<i>Metarhizium brunneum</i> H.J.S. 1154	2.3×10^7	16/07
		<i>M. brunneum</i> H.J.S. 1868	2.3×10^7	16/07
		<i>Beauveria bassiana</i> H.J.S. 1174	2.3×10^7	16/07
		<i>Clonostachys solani</i> H.J.S. 1828	2.3×10^7	16/07
		<i>Trichoderma atroviride</i> H.J.S. 1873	2.3×10^7	16/07
		<i>T. koningiopsis</i> H.J.S. 1874	2.3×10^7	16/07
		<i>T. gamsii</i> H.J.S. 1876	2.3×10^7	16/07
		Naturalis	2.3×10^7	16/07
	25/07 2013	Control	–	No treatment
		<i>M. brunneum</i> H.J.S. 1154	2.3×10^7	24/07
		<i>M. brunneum</i> H.J.S. 1868	2.3×10^7	24/07
		<i>Beauveria bassiana</i> H.J.S. 1174	2.3×10^7	24/07
		<i>Clonostachys solani</i> H.J.S. 1828	2.3×10^7	24/07
		<i>Trichoderma atroviride</i> H.J.S. 1873	2.3×10^7	24/07
		<i>T. koningiopsis</i> H.J.S. 1874	2.3×10^7	24/07
		<i>T. gamsii</i> H.J.S. 1876	2.3×10^7	24/07
		Naturalis	2.3×10^7	24/07

[†]Insecticide dose is expressed as mass of active ingredient per plantlet in the case of drenching, or per hectare in the case of spraying. Fungal treatments are expressed as number of conidia per plantlet.

dose: 500 kg/ha) was applied at first hoeing only in 2013. The negative control consisted of mock-treated plants wetted with 2 mL of tap water. Due to the high pressures of cabbage flea beetles (*Phyllotreta* spp.), experimental setup 1 was treated with insecticide Fastac 100 EC (a.i. α cypermethrin, 10% w/w; BASF) at a rate of 75 mL/ha (7.5 g a.i./ha) on 21/08/2012, 05/08/2013, and 15/07/2014 (Table 1). Different treatments were tested along two adjacent rows of 200 plants per row, planted by a mechanical transplanter. Accordingly, an area of 120 m² was tested for each treatment in Tržič in 2012 and 2013, and one of 105 m² in Šentjakob in 2014; these combinations of year, experimental site and cauliflower cultivar are further referred to as “experimental blocks.” Two rows were left untreated between treatments to prevent cross

contamination and drift. In all treatments, four plots of 2×10 plants each were marked for evaluation. In the Tržič experiment, 20 adjacent plants in each plot were dug out at harvest time for counting CRF pupae, root quality class evaluations (see below) and plant weight assessments, whereas in Šentjakob, 10 plants per plot were examined.

Experimental setup 2 for evaluating biopesticide agents

Each plant was inoculated by pipetting 2 mL of 1.15×10^7 viable conidia/mL prepared in 0.05% Tween 80. The tested fungi were *Metarhizium brunneum* (strain H.J.S. 1154; cited in Razinger *et al.*, 2014b as *M. anisopliae* 1154), *M. brunneum* (1868), *Beauveria bassiana* (1174),

Clonostachys solani (1828), *Trichoderma atroviride* (1873), *T. koningiopsis* (1874), and *T. gamsii* (1876). Conidial suspensions were prepared as described elsewhere (Razinger et al., 2014a). Plants in negative controls were drenched with 2 mL 0.05% Tween 80. The formulated biopesticide Naturalis (Biogard, CBC Europe; including *B. bassiana* strain ATCC 74040) was applied by mixing one part product with one part tap water and applying two mL per plant (Table 1). The experiment was performed in four replicates in 2012 and five in 2013. Each replicate consisted of 20 plants. The replicates were planted randomly in four or five parallel rows. In each replicate three root systems were dug out at harvest time for counting CRF pupae and root quality class evaluations (see below).

Evaluation at harvest

Cauliflower was harvested at 82–98 d after transplanting. Individual plants were cut 3–5 cm above soil level. Aboveground parts of harvested plants were weighed in the field using a portable scale (AXISBA3MK, Tehnica Železniki, Slovenia). Root systems of individual harvested plants were dug out and washed and graded into four root quality classes (QCL0: no CRF damage, QCL1: slight damage, QCL2: moderate damage, QCL3: severe damage). Examples of each class are illustrated in supporting information. The number of CRF pupae was determined by a flotation technique for samples of five (setup 1) or three (setup 2) root systems. The root quality class values of five (setup 1) or three plants (setup 2) were averaged and treated as a continuous variable.

Data analysis

Results from experimental setups 1 and 2 were analysed using mixed linear models within the REML framework. Initial models contained a fixed factor describing the experimental treatments and two random factors describing the experimental block (see methods above) and replicate identity to control for pseudo-replication. The significance of fixed factors was tested with *F* tests. In addition, we included the interaction experimental treatment $\times$ experimental block in order to account for variations of treatment effects between years, experiment sites and cauliflower cultivar. The interaction was treated as a random factor and its significance estimated by comparing the difference of the -2 log-likelihood of the models with and without the factor to a chi-square distribution with 1 degree of freedom (Pinheiro & Bates, 2000). Numbers of pupae per plant and plant weight were treated as continuous and log-transformed ($\text{Log}(x + 1)$ or $\text{Log}(x)$,

respectively). The resulting variables hence complied with the homoscedasticity and residual normality assumptions of the linear model framework. Root quality class data was not transformed and was treated as a continuous variable. It should be noted that it was not possible to treat the numbers of pupae per plant as count data (e.g., using a Poisson distribution) or root quality as an ordinal variable, since values from several plants were averaged during sample processing. A power analysis was carried out to understand the lack of significant differences among treatments of the experimental setup 2. Analyses were carried out with the statistical software JMP 12.1.0. (SAS Institute, Inc.).

Results

The CRF pressure varied considerably during the years 2011–2014. Number of eggs was the highest in 2011, when the peak of eggs laid was 16.8 ± 1.9 eggs/trap/d as recorded on 28/04/2011. The CRF appeared to have three generations in 2011. The second egg-deposition peak was observed on 5/07/2011, and the third on 9/09/2011. CRF egg-deposition peaked on 29/05/2012 (2.5 ± 0.5 eggs/trap/d) and 20/05/2013 (1.7 ± 0.5 eggs/trap/d) but deposition trends and CRF flights were not as clear as in 2011. In 2014, two CRF generations were observed with flight peaks on June 20, 2014 (1.6 ± 0.5 eggs/trap/d) and August 7, 2014 (0.4 ± 0.2 eggs/trap/d) (Fig. 1). The CRF egg deposition was patchy, as indicated by the large coefficients of variation of CRF egg counts; the degree of patchiness also exhibited substantial interannual variation. The average (and minimum–maximum) coefficient of variation of eggs counted in felt traps was 39.0% (0.1%–218.4%) in 2011, 12.6% (0.1%–58.4%) in 2012,

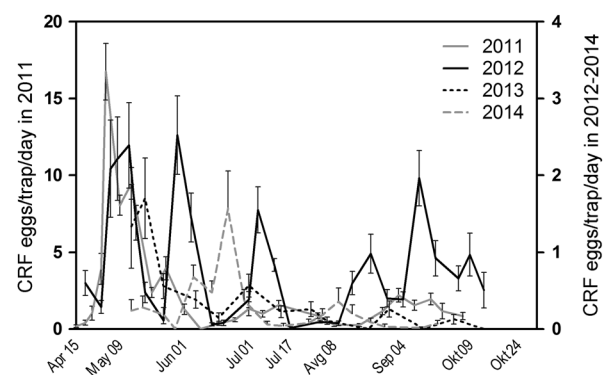


Fig. 1 CRF egg laying in field experiments in 2011–2014. CRF egg laying was monitored using felt traps. The dates of field experiments were 17/07/2012, 25/07/2013, and 26/07/2014. Data are presented as means $\pm$ SE ($n = 20$).

4.0% (0.0%–22.3%) in 2013 and 1.6% (0.0%–8.1%) in 2014.

The treatments had a significant effect on the number of recovered CRF pupae in experimental setup 1 ($F_{4,32} = 39.07$; $P < 0.0001$) but not in experimental setup 2 ($F_{8,67} = 1.71$; $P = 0.113$). Interactions between treatments and experimental blocks were not significant. *Post hoc* tests on experimental setup 1 revealed that numbers of CRF pupae were reduced by spinosad treatment (mean of 1.01 ± 0.13 CRF pupae per plant; 19.8% normalized to control) and increased with nitrogen lime application (5.90 ± 0.66 ; 144.5% control). The highest but nonsignificant reduction of the number of CRF pupae in experimental setup 2 were recovered in roots treated with *Naturalis* (2.78 ± 0.71 ; 46.5% control) and *M. brunneum* H.J.S. 1868 (3.38 ± 0.86 ; 56.6% control; Fig. 2). A power analysis for the noncommercial fungi revealed a chance of 69%, that the treatments had different effects and that the experiments in setup 2 should have been about 2.5 times larger to detect differences among fungal strains (minimal sample size estimated = 177, actual sample size = 68).

In experimental setup 1, the effects of pesticide treatments on root quality varied between experimental blocks (interaction treatment $\times$ experimental block: $\chi^2_1 = 7.1$; $P = 0.008$). Despite these variations, *post hoc* tests revealed that spinosad and nitrogen lime reduced root damage relative to controls (Fig. 2). In experimental setup 2, fungal treatments had no significant effect on root quality ($F_{8,32} = 0.797$; $P = 0.609$). A power analysis for the noncommercial fungi revealed a chance of 81% that the treatments had different effects and that the experiments in setup 2 should have been about four times larger to detect differences among fungal strains (minimal sample size estimated = 139, actual sample size = 36).

Treatments affected plant weight in the experimental setup 1 ($F_{4,33} = 5.62$; $P = 0.001$) but not in experimental setup 2 ($F_{8,86} = 0.290$; $P = 0.968$). Interactions between treatments and experimental block were not significant. *Post hoc* tests of the results of setup 1 revealed that nitrogen lime and λ -cyhalothrin increased plant weight relative to controls (Fig. 2). A power analysis of the experimental setup 2 revealed that chances the treatments were actually different were 42%, and that the experiments in setup 2 should have been about twice larger to detect differences among fungal strains (minimal sample size estimated = 1581, actual sample size = 757).

Discussion

The monitoring results showed a strong interannual variation in CRF population dynamics. The peak intensity of

CRF egg laying ranged from 16.8 ± 1.9 in 2011 to 1.6 ± 0.5 eggs/trap/d in 2014. Additionally, the laid eggs' distribution was patchy. The average annual coefficient of variation of the eggs counted on felt traps ranged from 39.0% in 2011 to 1.6% in 2014. Similar to our results, the high spatiotemporal variability of inter- and intra-annual CRF infestation was also observed by Dalthorp and Dreves (2008). This information demonstrates that a detailed monitoring programme of the pest is a prerequisite during development of a holistic CRF IPM strategy.

Spinosad protected cauliflower roots by reducing the number of CRF pupae and root damage significantly without affecting aboveground plant biomass. Nitrogen lime increased significantly the number of pupae in the root systems; however, it improved the root quality and also resulted in a significant increase of aboveground plant biomass. The nonformulated entomopathogenic and other fungi tested in field experiments caused a nonsignificant reduction in the number of CRF pupae and improvement of root quality. Plant weight (i.e., yield) was, however, indistinguishable from that of controls. The nonformulated entomopathogenic and other fungi were selected for the experiments because of their ability to colonize brassica roots (Razinger *et al.*, 2014b) and concomitant pathogenicity to CRF (Razinger *et al.*, 2014a).

Instead of wide-scale spraying or soil-incorporation of the pesticides, single plant treatment was preferred, where possible. Drenching young plants with conidial suspensions before transplanting may allow the establishment of mycelial growth for example of rhizosphere competent biopesticide agents on or near the roots. The strategy could present a less costly and more environmental friendly way to disseminate biopesticides, as it avoids repeated sprayings or broad scale applications. Drenching roots with spinosad gave the best CRF control in field experiments and ensured long-term root protection based on a single treatment. However, this did not result in a change of plant weights, which might possibly be explained by the interactions of alpha cypermethrin and other insecticides evaluated in experimental setup 1. Neither aerial application of λ -cyhalothrin nor a drenching application of thiamethoxam reduced the number of CRF pupae or degree of root damage. This result is in contrast to that reported by Joseph and Zarate (2015) in which a λ -cyhalothrin outperformed spinosad in CRF control. Λ -cyhalothrin significantly increased the average aboveground plant weight, perhaps because it successfully controlled any other occurring cauliflower pests, including *Contarinia nasturtii*, *Phyllotreta* spp., and *Pieris* spp. Nitrogen lime had no measurable toxic effects on CRF eggs and larvae, even higher numbers of pupae being observed

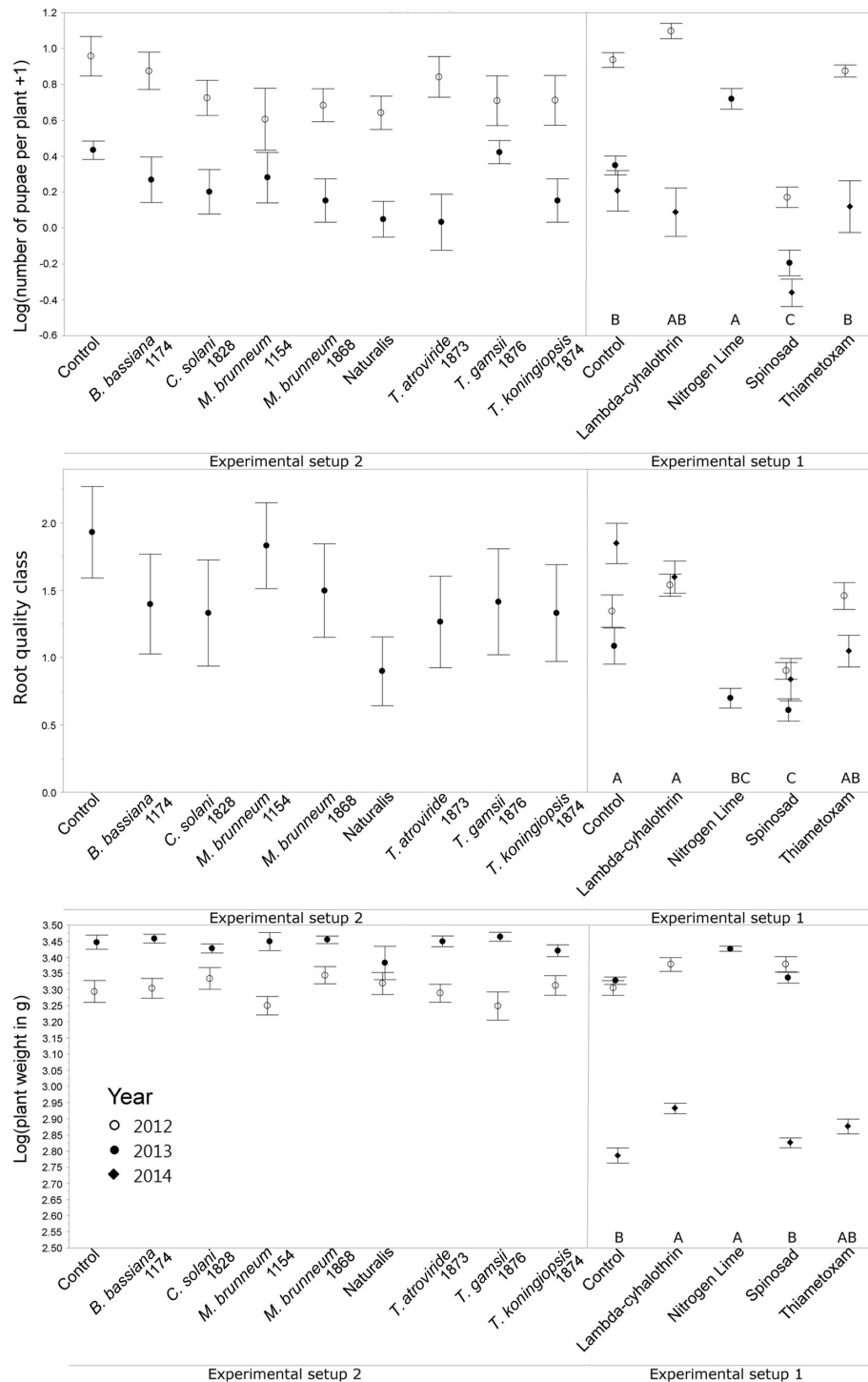


Fig. 2 Effects of insecticides (experimental setup 1) and fungal treatments (setup 2) on CRF pupae in root samples, cauliflower CRF-related root damage, and aboveground plant weight. Data presented are means $\pm$ SE. CRF pupae and root quality class error bars were calculated from 16 samples in 2012 and 2013, 8 in 2014 (setup 1), 4 in 2012, and 5 in 2013 (setup 2). Plant weight error bars were calculated from 80 samples 2012–2014 (setup 1) and 80 in 2012 or 20 in 2013 (setup 2). Treatments marked by various letters differ significantly. Naturalis, a bioinsecticide based on *Beauveria bassiana* ATCC 70404; Spinosad, the active ingredient of insecticide Laser; λ -cyhalothrin, the active ingredient of insecticide Karate Zeon; Thiamethoxam, the active ingredient of insecticide Actara. Root quality classes in experimental setup 2 were evaluated only in 2013.

in the control. It did, however, increase aboveground plant weight significantly and improve root quality, which perhaps relates to its nitrogen fertilization effect, ensuring overall plant vigour. Apparently because of lacking statistical power fungal treatments had no significant influence on the number of CRF pupae recovered from cauliflower roots. Treatment with the tested fungi, however, resulted in nonsignificant improvement of root quality. Similarly, using *M. anisopliae* 1154 and Naturalis resulted in an insignificant CRF reduction and root quality improvement in experiments performed on experimental stations in broccoli, with yields statistically indistinguishable from the spinosad treatment (Herbst *et al.*, 2017). The authors suggested that beneficial fungi could effectively control CRF in organic brassica such as cauliflower, broccoli, and brussels sprouts, where the produce is not directly attacked by CRF, but for radish, swedes, or turnips, one should consider treatments that have a greater impact on CRF (e.g., spinosad; Herbst *et al.*, 2017). This study increased the diversity of fungal taxa evaluated for CRF control compared to the Herbst *et al.* (2017) study. The field results indicate that the three *Trichoderma* species and *C. solani* (H.J.S. 1828) tested in this study offered a level of protection of cauliflower roots similar to that afforded by entomopathogenic *M. brunneum* isolates (H.J.S. 1154 and 1868) and *B. bassiana* (ATCC 70404; bioinsecticide Naturalis).

Attempts to control CRF with fungi have mostly been performed in the laboratory (e.g., Vänninen *et al.*, 1999; Bruck *et al.*, 2005; Razinger *et al.*, 2014a; Myrand *et al.*, 2015) or glasshouse (e.g., Vänninen *et al.*, 1999; Chandler & Davidson, 2005; Razinger *et al.*, 2014b). According to our knowledge, only Chandler and Davidson (2005) have performed field experiments, using *Metarhizium anisopliae* to control CRF. While fungi applied in these studies performed well in glasshouse experiments, no control was achieved in the field, even with much greater conidial numbers than those used in this study. Chandler and Davidson (2005) explained their finding by assuming that their inoculum did not reach the roots in drenching applications. Vänninen *et al.* (1999), after achieving insignificant CRF control in glasshouse experiments, suggested focusing on finding isolates showing high virulence against specifically targeted pests, or on the inundative control of pupae to achieve a long-term reduction of pest population in a given area. Instead of drenching, as applied by Vänninen *et al.* (1999), we focused on utilizing the fungal isolates' rhizosphere competence in order to protect the transplants against belowground CRF maggots.

The field experiments showed that a single drench of cauliflower plantlets with spinosad offered consistent

and enduring CRF control. Treatment with commercial *B. bassiana* ATCC 74040 (Naturalis) and other nonformulated fungi did not significantly decrease CRF pressure in field experiments. Thus, the proposed biological control strategy involving a single cauliflower plantlet inoculation with beneficial fungi needs to be improved before it can be considered a functional CRF control strategy. More fungal applications and cooperation with formulation specialists might improve such a strategy, as suggested also by Herbst *et al.* (2017). Further, when utilizing rhizosphere competence of insect pathogenic or other fungi for soil pest control, preceding experiments should separately aim at evaluating (i) insect pathogenicity and (ii) rhizosphere competence. Such a sequential approach could narrow down the list of potentially beneficial fungal candidates exhibiting both, insect pathogenicity and rhizosphere competence, and thus enable an increase of the number of replicates and the consequent statistical power of the experiments. What is more, such an approach might broaden the repertoire of beneficial fungi used for CRF and other soil pest management to include fungi also from nonentomopathogenic genera.

Acknowledgments

We are grateful to Anton Aljančič from Kovor and Andrej Mrčun from Šentjakob for sharing their cauliflower expertise and their fields for this study. We are grateful to Tanja Kokalj, Aleksandra Podboj-Ronta, and Marko Mechora for their technical assistance. We are also grateful for the constructive comments of the reviewers and the associate editor of the special issue. The research was realized within the frame of FP7 Projects CropSustaIn, grant agreement FP7-REGPOT-CT2012-316205 and PURE, grant agreement 2658652012–2013, COST Action FA1405 Crop Arthropod Microbe Interactions, a Sciex-NMSch scholarship to J. Razinger (11.090, acronym RhizoShield), the Agrobiodiversity program group (P4-0072) of the Agricultural Institute of Slovenia and a grant from the ARRS Slovenian Science Foundation to H.J. Schroers (J4-5527).

Disclosure

This article reports the results of research only. Mention of trade names or commercial products in this publication is solely for the purpose of providing specific information and does not imply recommendation or endorsement by the Agricultural Institute of Slovenia, Zurich University of Applied Sciences, Agroscope or INRA.

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Manuscript received October 28, 2016

Final version received August 21, 2017

Accepted August 26, 2017

Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

Fig. S1 Graphical representation of the different root quality classes which was used as a visual scale for categorizing field-collected cauliflower root systems into appropriate root quality classes.